



Faculty of Computer Applications

BCA Sem-1 (AY-2026-27)

Lab Book

Sub Code : 05BC3103

(Programming Practices : 1)

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GitHub link : <https://github.com/dhruvikkiyada3456/BCA-1-pp-1--92600527011-Dhruvik>

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Index

Unit : 1	
Program No	Definition
1	Print "hello world".
2	2 Values from user and Perform All Arithmetic Operations
3	Print numbers from 1 to 20.
4	Input 3 values from user and find out its Average.
5	Calculate the Square and Cube of a number.
6	Interchange the value of two numbers
7	Find largest out of three numbers.
8	Find out Fibonacci series up to given n values.
9	Find out Number is Odd or Even.
10	Find out sum of digits of a given number.

Unit : 2	
Program No	Definition

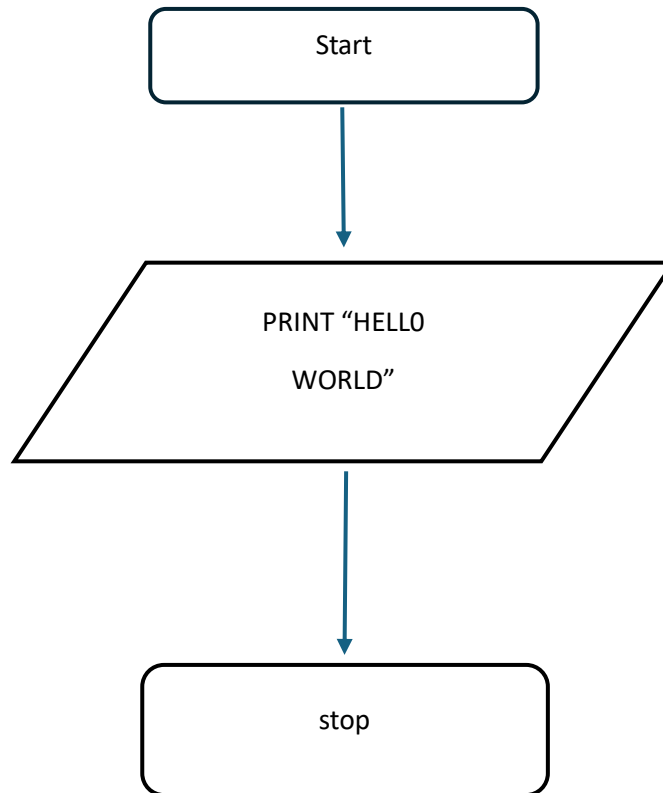
1.	Write a Program to Print "Hello World".
2.	Write a Program that takes 2 Values from user and Perform All Arithmetic Operation.
3.	Write a program that takes 3 values for Principle Amount, Rate of Interest and No. of Years and find out Simple Interest.
4.	Write a program that input value from user and find out Area of Circle.
5.	Write a program that input two values from user and find out Area of Rectangle.
6.	Write a program that input 3 values from user and find out its Average.
7.	Write a program to Calculate the Square and Cube of a number.
8.	Write a program Interchange the value of two numbers using third variable.
9.	Write a program Interchange the value of two numbers without using third variable.
10.	Write a program that input quantity, price, discount and display final amount.

(1) Print “hello world”.

Algorithm:

step 1: start
step 2 : PRINT”hello world”
step 3: stop

flowchart:



Code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    clrscr();
    printf("hello world");
    getch();
}
```

Output:



```
hello world
```

(2) 2 Values from user and Perform All Arithmetic Operations.

Algorithm:

Step 1: start

Step 2: DECLARE variables $x=0.0, y=0.0, a, s, m, d$

Step 3: INPUT x, y

Step 4: $a=x+y$

Step 5: $s=x-y$

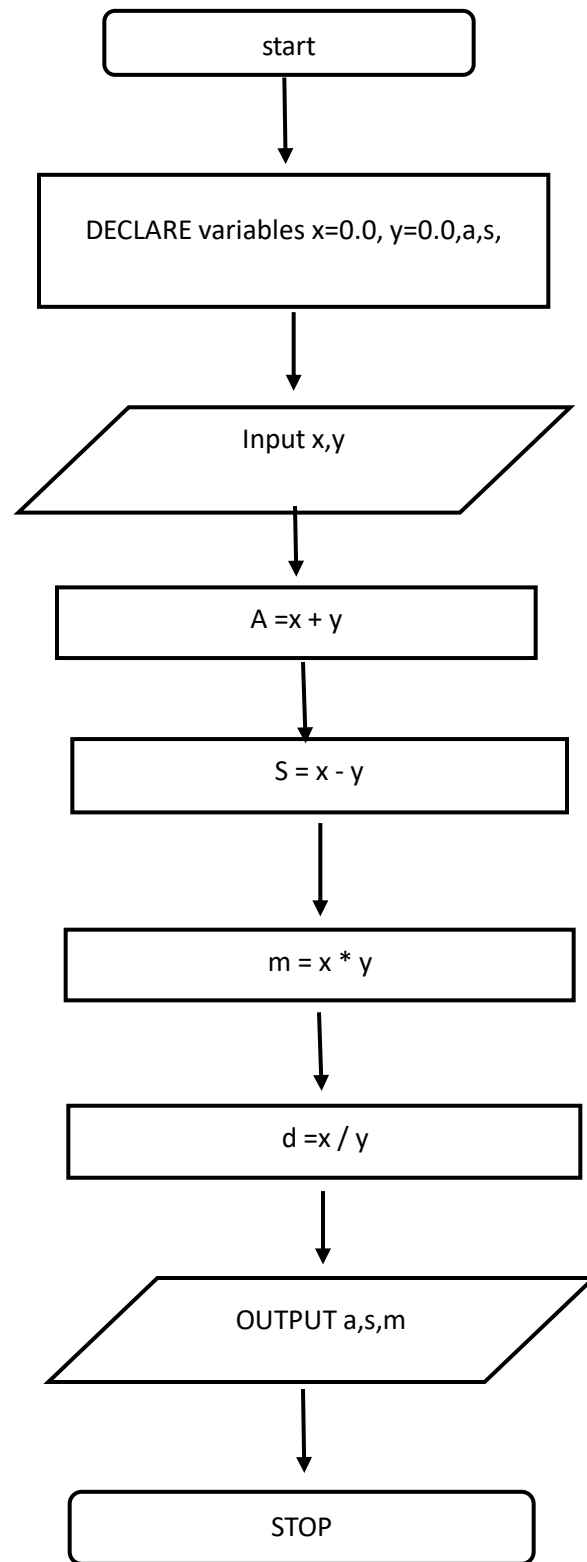
Step 6: $m= x*y$

Step 7: $d=x/y$

Step 8: OUTPUT a, s, m, d

Step 9: stop

Flow chart:



Code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a,b;
    clrscr();

    printf("enter first number:");
    scanf("%d",&a);
    printf("enter second number:");
    scanf("%d",&b);

    printf("addition:   %d\n",a+b);
    printf("subtraction: %d\n",a-b);
    printf("multiplication:%d\n",a*b);
    printf("divison:   %d\n",a/b);
    printf("remainder:  %d\n",a%b);
    getch();
}
```

Output:

```
enter first number:10
enter second number:5

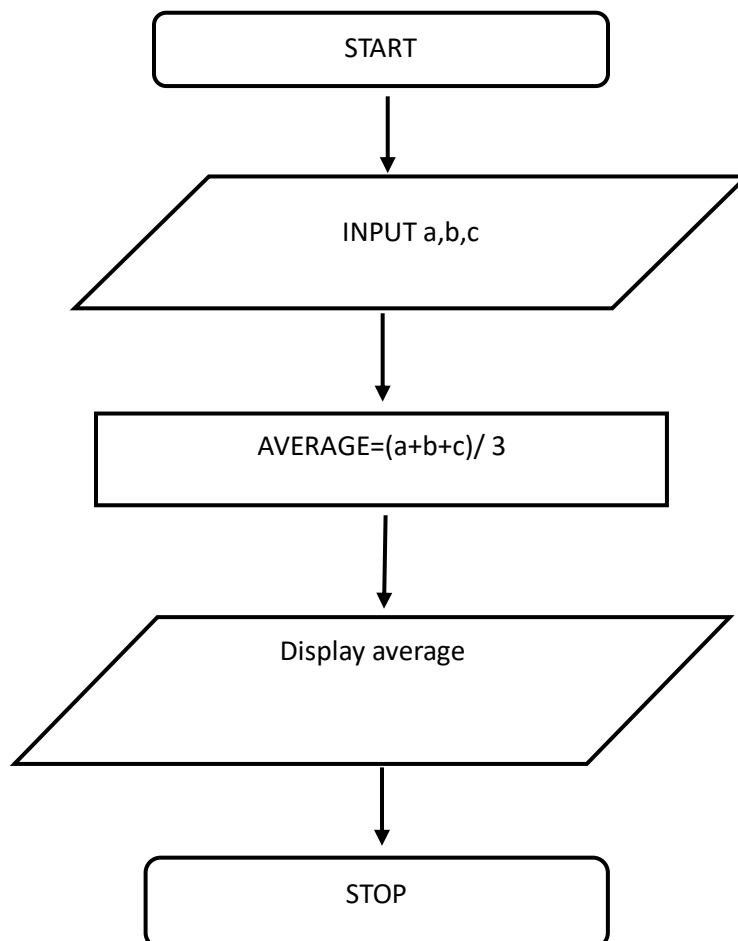
sum = 15.000000
substrection = 5.000000
multiplication = 50.000000
divison = 2.000000_
```

(4) Input 3 values from user and find out its Average.

Algorithm:

- **Step 1:** Start.
- **Step 2:** Declare float variables num1, num2, num3, sum, aur avg.
- **Step 3:** Read/Input three numbers from the user and store them in num1, num2, and num3.
- **Step 4:** Calculate the sum of three numbers: $\text{sum} = \text{num1} + \text{num2} + \text{num3}$.
- **Step 5:** Calculate the average: $\text{avg} = \text{sum} / 3.0$.
- **Step 6:** Print/Display the calculated value of avg.
- **Step 7:** Stop.

Flowcharts:



Code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    float num1,num2,num3,average;
    clrscr();
    printf("enter three numbers:");
    scanf("%f %f %f",&num1, &num2, &num3);

    average = (num1 + num2 + num3) / 3;

    printf("\n average %f",average);
    getch();
}
```

Output:

```
enter three numbers:10
50
80
average: 46.666668
```

(5) Calculate the Square and Cube of a number.

Algorithm:

Step 1: START

Step 2: DECLARE VARIABLE x, s, c

Step 3: PRINT "enter any number"

Step 4: INPUT x

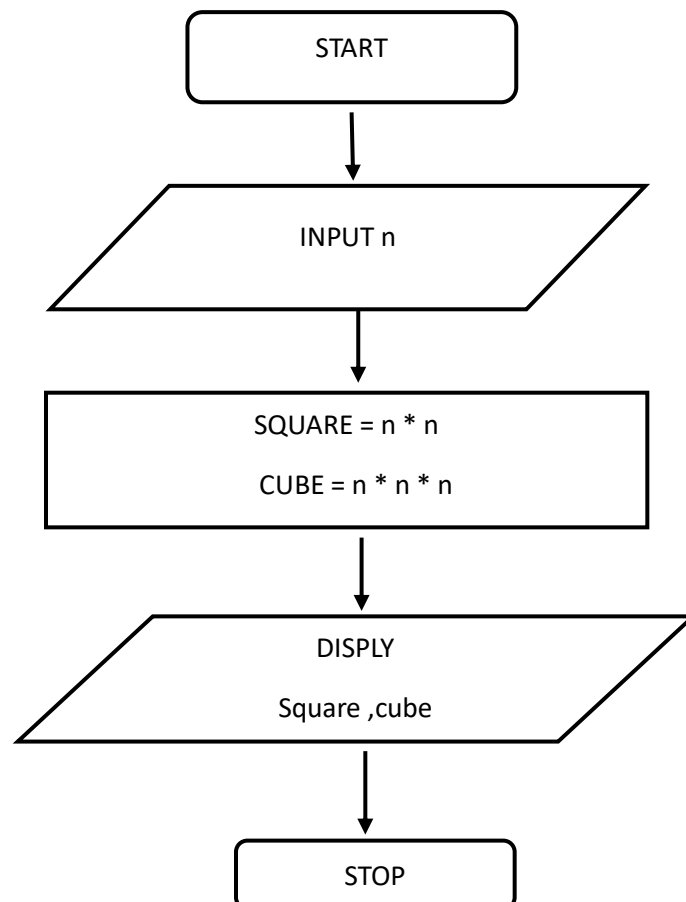
Step 5: $s = x * x$

Step 6: $c = x * x * x$

Step 7: OUTPUT s, c

Step 8: STOP

Flowchart:



Code:

```
#include<stdio.h>

#include<conio.h>

void main()
{
    int x,y,z;
    clrscr();
    printf("\n enter value of x");
    scanf("%d",&x);

    y=x*x;
    printf("\n square is :%d",y);

    z=x * x * x;
    printf("\n cube is %d",z);
    getch();
}
```

Output:

```
number : 45  
square : 2025  
cube : 25589
```

(6) Interchange the value of two numbers.

Algorithm:

Step 1: START

Step 2: DECLARE VARIABLE a,b,temp

Step 3: INPUT VALUE of a,b

Step 4 : temp = a

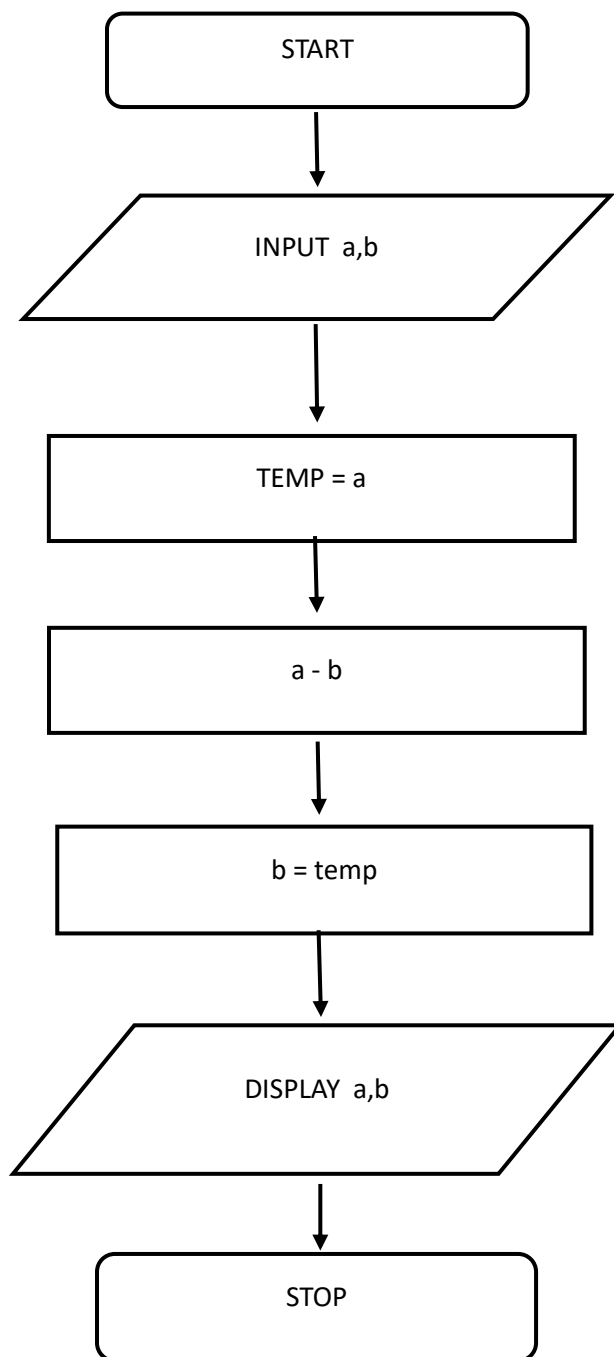
Step 5 : a = b

Step 6 : b = temp

Step 7 : PRINT VALUE of a and b

Step 8 : STOP

Flowchart:



Code:

```
#include<stdio.h>

#include<conio.h>

void main()
{
    int x,y,z;
    clrscr();
    printf("enter value of x:");
    scanf("%d",&x);

    printf("enter value of y:");
    scanf("%d",&y);
    printf("\n before interchange: x=%d y=%d",x,y);

    z=x;
    x=y;
    y=z;
    printf("\n After interchange : x=%d y=%d",x,y);
    getch();
}
```

Output:

```
enter value of a:45
enter value of b:50

Before interchange: a=45, b=50
after interchange: a=50, b=45_
```

(8) Find out Fibonacci series up to given n values.

Algorithm:

Step 1: START

Step 2: DECLARE VARIABLE n , $t_1 = 0$, $t_2 = 1$, $i = 1$

Step 3: INPUT VALUE of n

Step 4: PRINT t_1

Step 5: calculate $= t_1 + t_2$

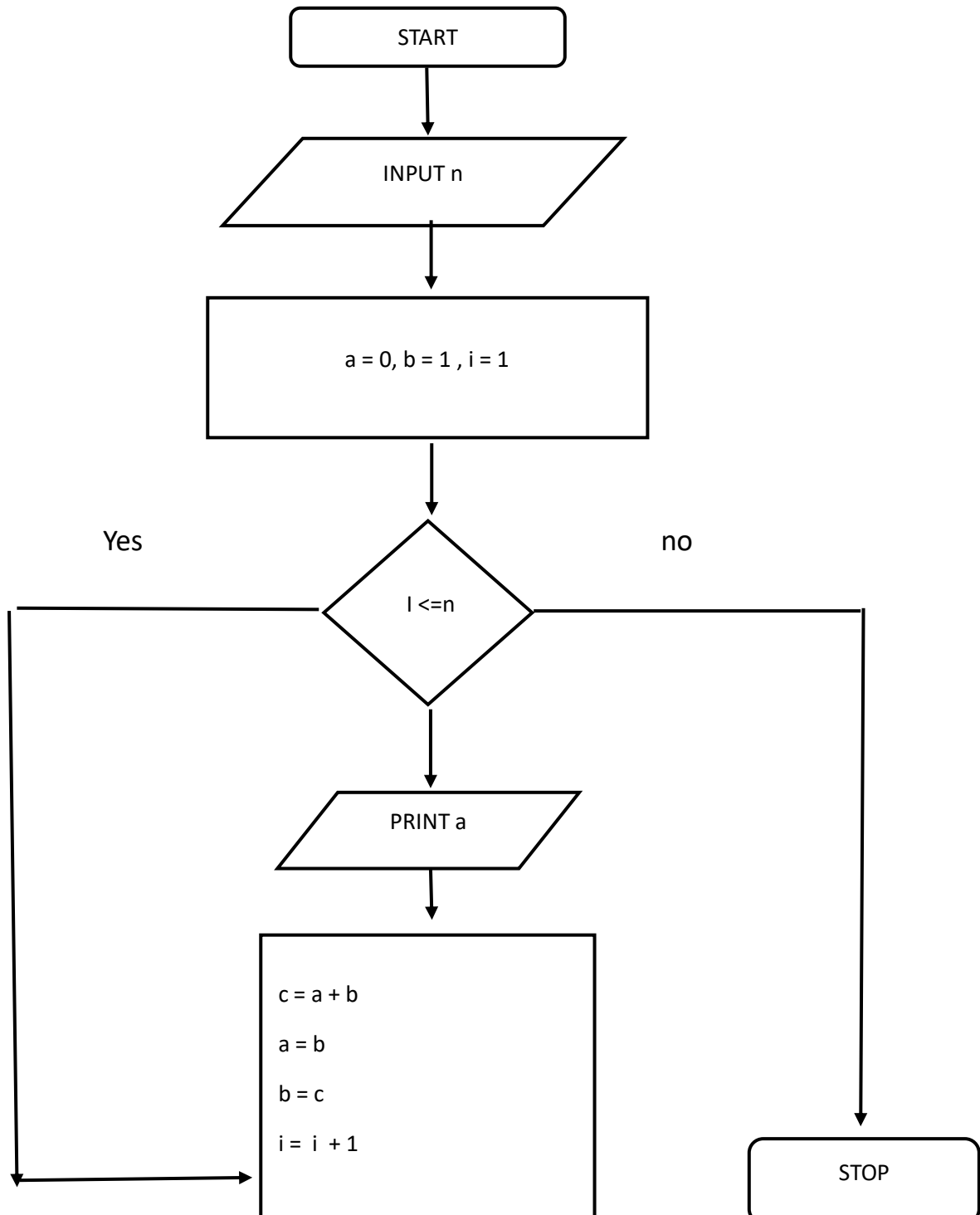
Step 6: $t_1 = t_2$ and $+ 2$

Step 7: $i = i + 1$

Step 8: IF $i \leq n$ then

Step 9: STOP

Flowchart:



Code:

```
#include <stdio.h>

int main() {
    int n, i;
    int a = 0, b = 1, c;

    printf("Enter number of terms: ");
    scanf("%d", &n);

    printf("Fibonacci Series: ");
    for (i = 1; i <= n; i++) {
        printf("%d ", a);
        c = a + b;
        a = b;
        b = c;
    }

    return 0;
}
```

Output:

```
enter number of terms:10
fibonacci series:0112358132134_
```

(9) Find out Number is Odd or Even.

Algorithm:

Step 1: START

step :2 DECLARE variable a

step 3 : PRINT “enter any number”

step 4: INPUT a

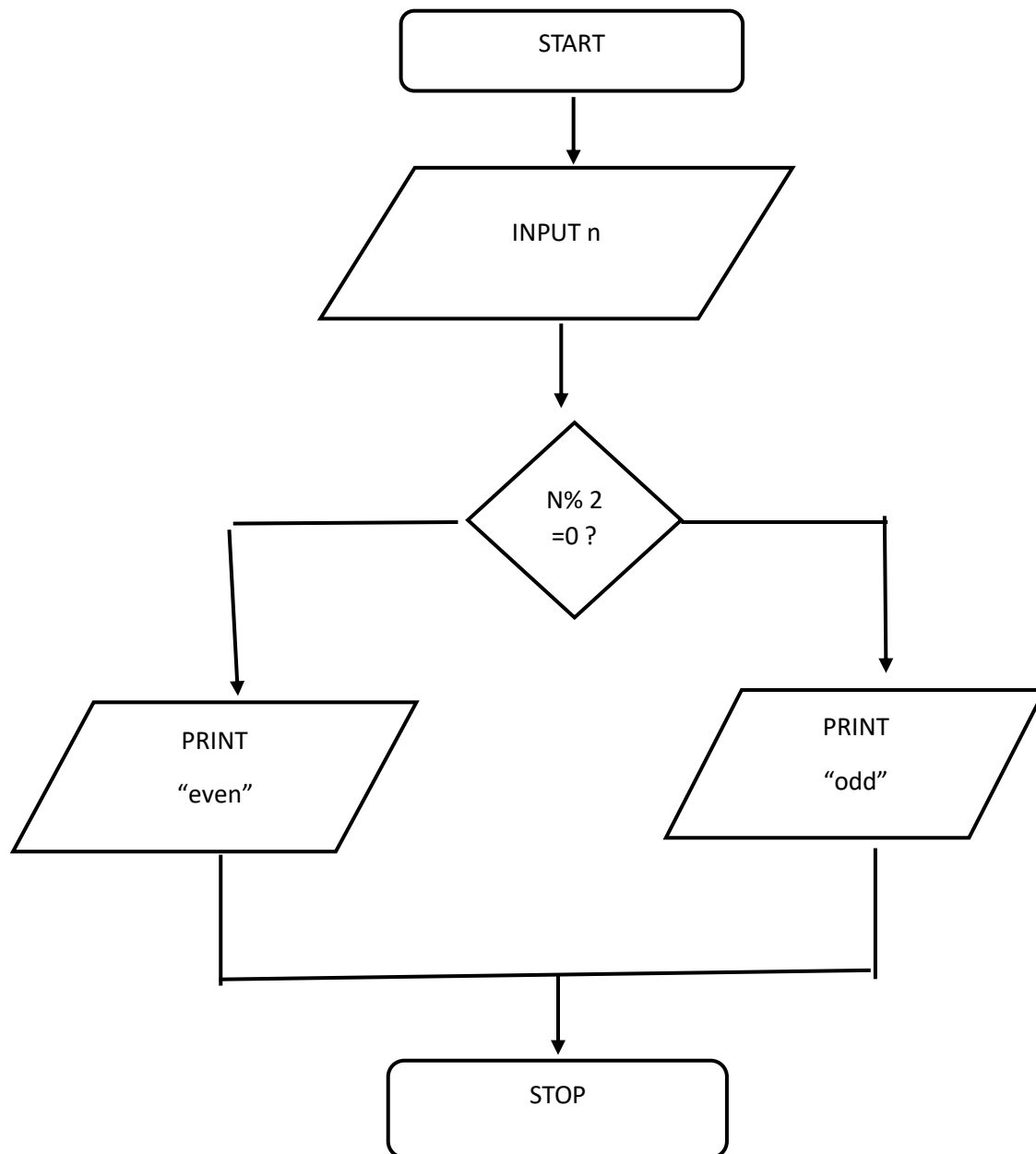
step 5 : if $A/2 == 0$

PRINT” Number is even”

Step 6: ELSE PRINT “number is odd”

Step 7 : STOP

Flowchart:



Code:

```
#include<stdio.h>

#include<conio.h>

void main()
{
    int x,y,z;
    clrscr();
    printf("\n enter any number:");
    scanf("%d",&x);
    y=x%2;
    if(y==0)
    {
        printf("\n number is even");
    }
    else
    {
        printf("\n number is odd");
    }
    getch();
}
```

Output:

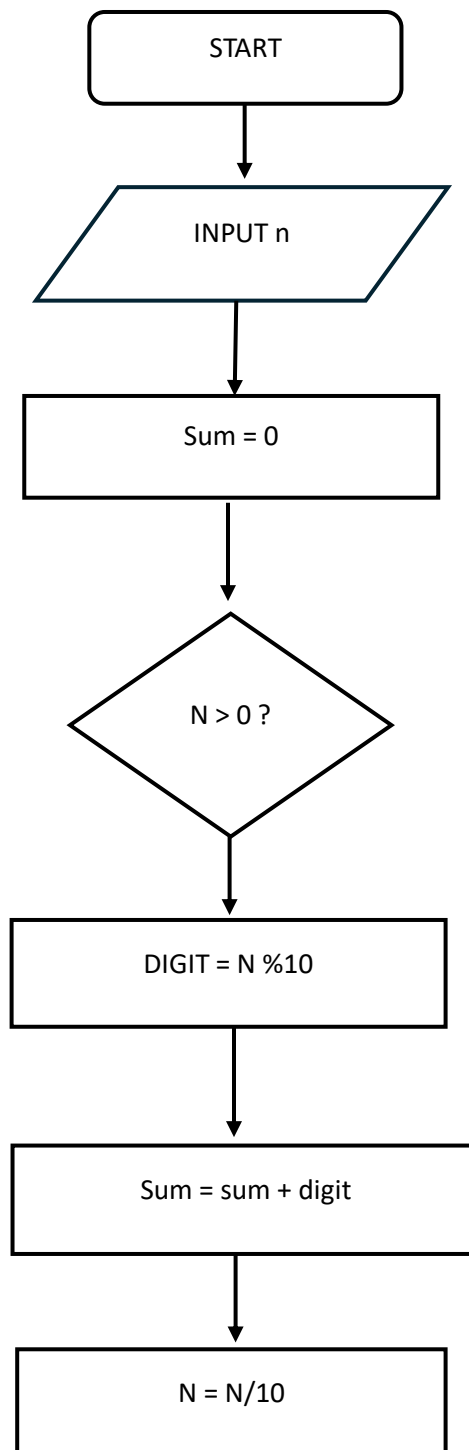
```
enter a number:4
4 is even:_
```

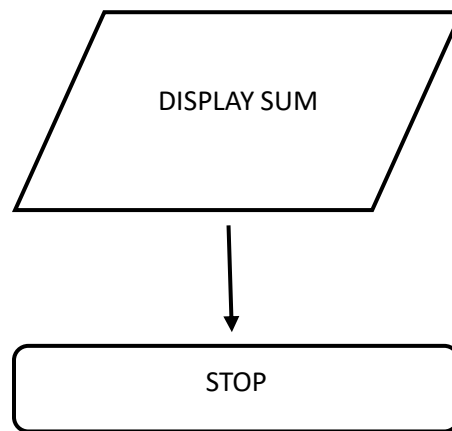
(10) Find out sum of digits of a given number.

Algorithm:

- **Step 1:** Start.
- **Step 2:** Take a number from the user as num.
- **Step 3:** Set sum = 0.
- **Step 4:** While num is greater than 0, repeat these steps:
 - $\text{sum} = \text{sum} + (\text{num} \% 10)$
 - $\text{num} = \text{num} / 10$
- **Step 5:** Print the sum.
- **Step 6:** Stop.

Flowchart:





Code:

```
#include <stdio.h>

int main() {
    int num, sum = 0;

    printf("Enter a number: ");
    scanf("%d", &num);

    while (num > 0) {
        sum += num % 10; // 1. Extract the last digit and add it to sum
        num /= 10;      // 2. Chop off the last digit
    }
    printf("Sum of digits = %d\n", sum);
    getch();
}
```

Output:

```
enter a number:45  
sum of digits=9_
```

UNIT : 2

(1) Write a Program to Print "Hello World".

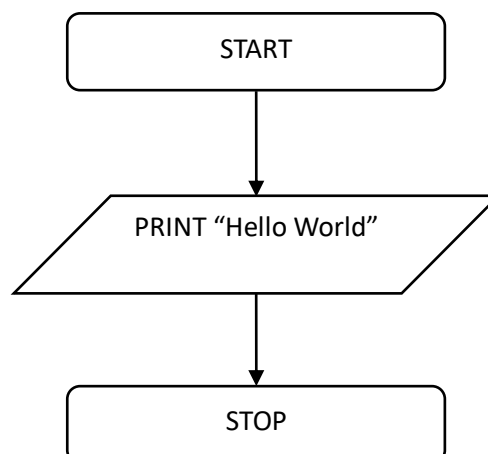
Algorithm :

Step 1 : START

Step 2 : PRINT "Hello World"

Step 3 : STOP

Flowchart :

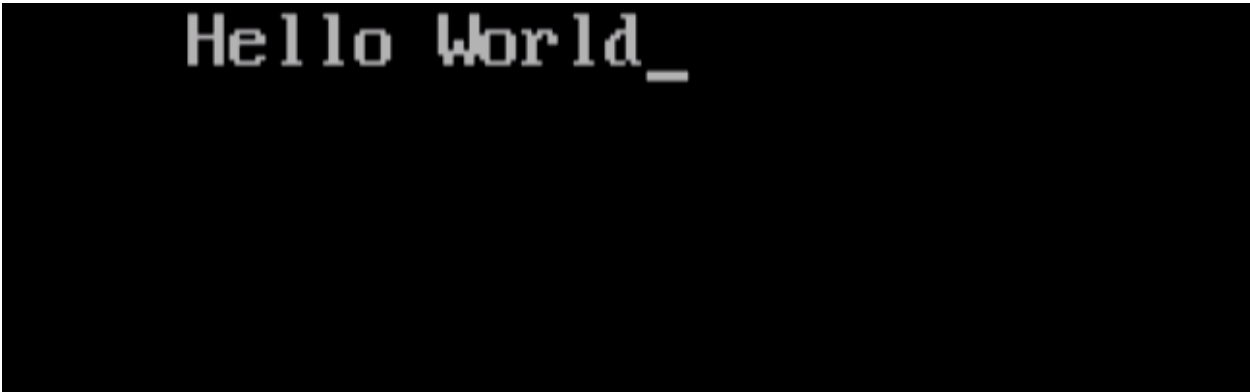


Code :

```
#include<stdio.h>
#include<conio.h>

void main(){
    clrscr();
    printf("Hello World");
    getch();
}
```

Output :

A screenshot of a terminal window with a black background. The text "Hello World_" is displayed in a light gray, monospaced font at the top of the window. The underscore at the end of the text indicates that the program is waiting for a character input, as shown by the getch() function in the code above.

Hello World_

(2) Write a Program that takes 2 Values from user and Perform All Arithmetic Operation.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables $x=0.0$, $y=0.0$, a, s, m, d

Step 3 : INPUT x,y

Step 4 : $a = x + y$

Step 5 : $s = x - y$

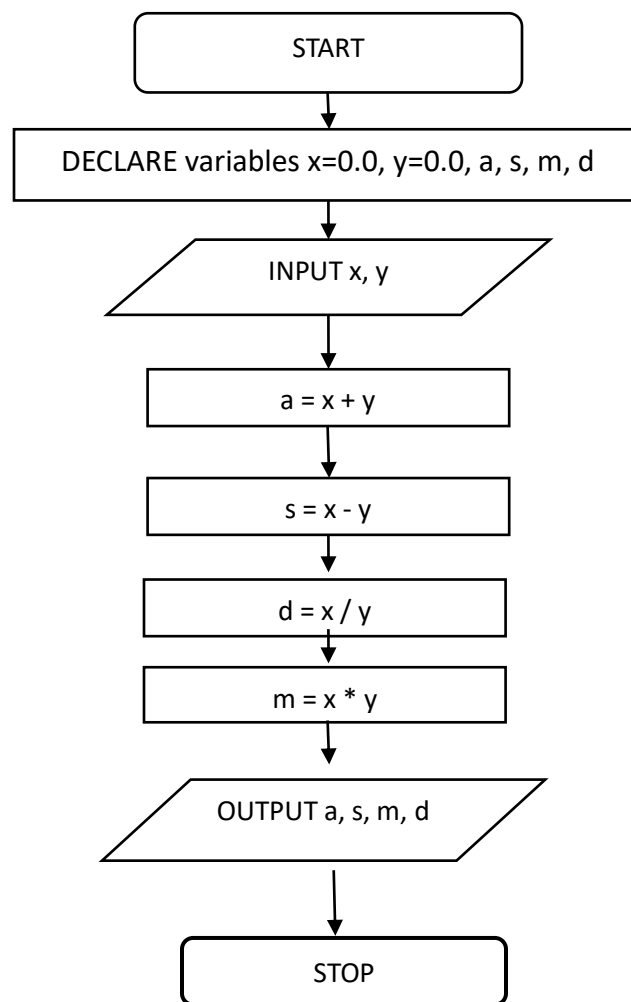
Step 6 : $m = x * y$

Step 7 : $d = x / y$

Step 8 : OUTPUT a, s, m, d

Step 9 : STOP

Flowchart :



Code :

```
#include<stdio.h>

#include<conio.h>

void main()
{
    float x,y;
    clrscr();
    printf("\n Enter Number 1: ");
    scanf("%f",&x);
    printf("\n Enter Number 2: ");
    scanf("%f",&y);

    printf("Addition Is: %.2f \n",x+y);
    printf("Subtraction Is: %.2f \n",x-y);
    printf("Multiplication Is: %.2f \n",x*y);
    printf("Division Is: %.2f \n",x/y);

    getch();
}
```

Output :

```
Enter Number 1: 23
```

```
Enter Number 2: 2
```

```
Addition is: 25.00
```

```
Subtraction is: 21.00
```

```
Multiplication is: 46.00
```

```
Division is: 11.50_
```

(3) Write a program that takes 3 values for Principle Amount, Rate of Interest and No. of Years and find out Simple Interest.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables SI, P, R, T

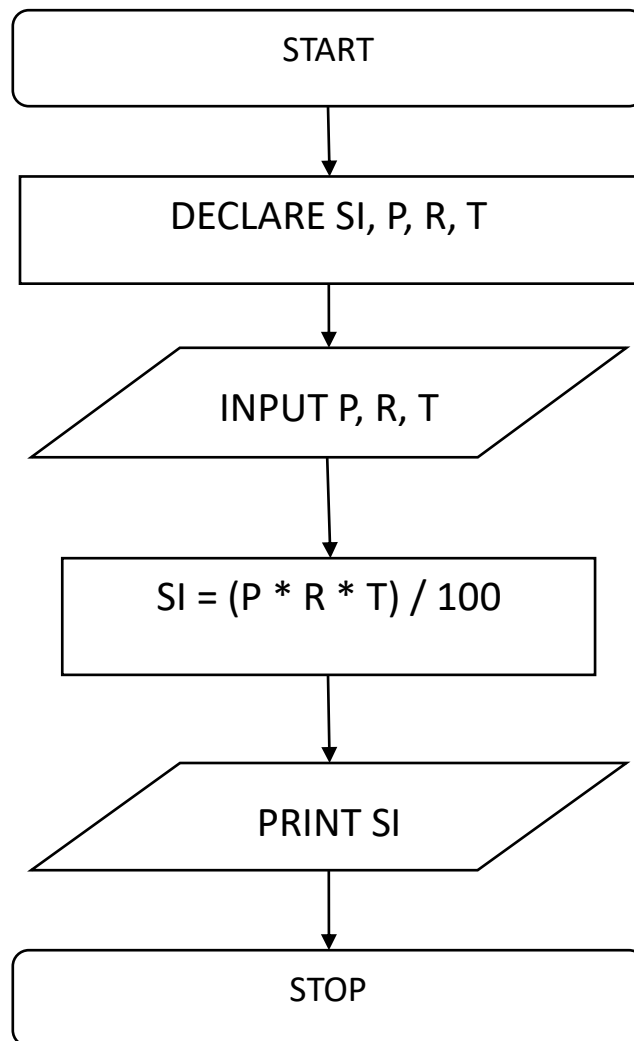
Step 3 : INPUT P, R, T

Step 4 : $SI = (P * R * T) / 100$

Step 5 : PRINT SI

Step 6 : STOP

Flowchart :



Code :

```
#include<stdio.h>

#include<conio.h>

void main(){

    float SI,P,R,T;

    clrscr();

    printf("Enter Principal Amount: \n");

    scanf("%f",&P);

    printf("Enter Rate Of Interest: \n");

    scanf("%f",&R);

    printf("Enter Total Time: \n");

    scanf("%f",&T);

    SI=(P*R*T)/100;

    printf("*****\n");

    printf("*Principal Amount=%.2f*\n",P);

    printf("*Rate of Interest=%.2f *\n",R);

    printf("*Time=%.2f *\n",T);

    printf("*****\n");

    printf("*Simple Interest=%.2f *\n",SI);

    printf("*****");
```

```
        getch();  
    }
```

Output :

```
Enter Principal Amount:  
500  
Enter Rate Of Interest:  
2.5  
Enter Total Time:  
5  
*****  
*Principal Amount=500.00*  
*Rate of Interest=2.50 *  
*Time=5.00              *  
*****  
*Simple Interest=62.50 *  
*****
```

(4) Write a program that input value from user and find out Area of Circle.

Algorithm :

Step 1 : START

Step 2 : DECLARE variable r, area

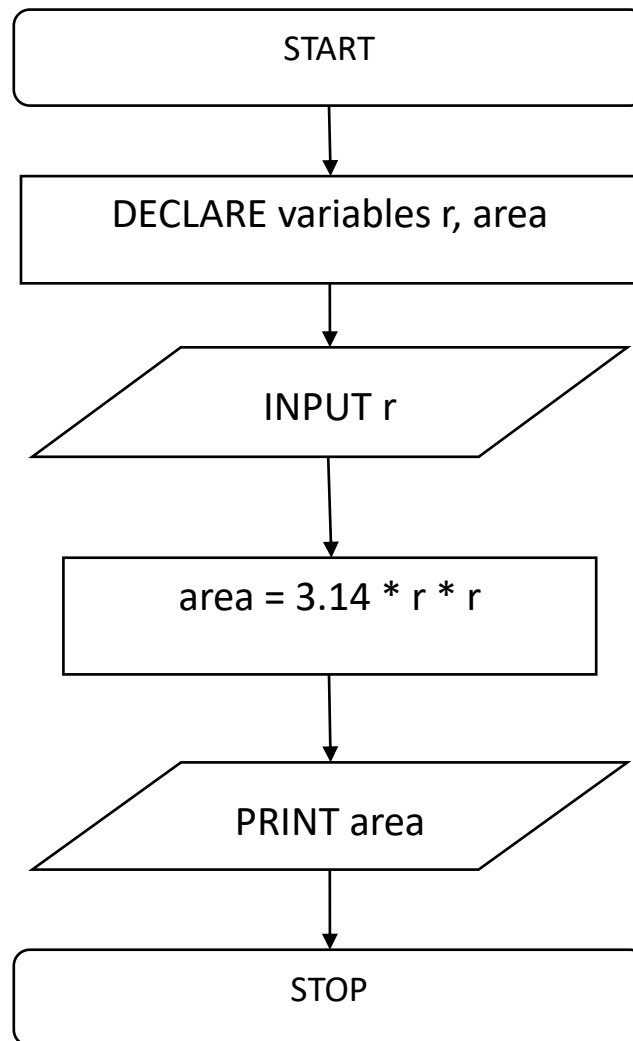
Step 3 : INPUT r

Step 4 : $\text{area} = 3.14 * r * r$

Step 5 : PRINT area

Step 6 : STOP

Flowchart :



Code :

```
#include<stdio.h>
#include<conio.h>

void main()
{
    float Radius;
    clrscr();
    printf("Enter Radius Of Circle: \n");
    scanf("%f",&Radius);
    printf("The Area Of Circle is: %f",3.14*Radius*Radius);
    getch();
}
```

Output :

```
Enter Radius Of Circle:  
3  
The Area Of Circle is: 28.260000_
```

(5) Write a program that input two values from user and find out Area of Rectangle.

Algorithm :

Step 1 : START

Step 2 : DECLARE variable l, b, area

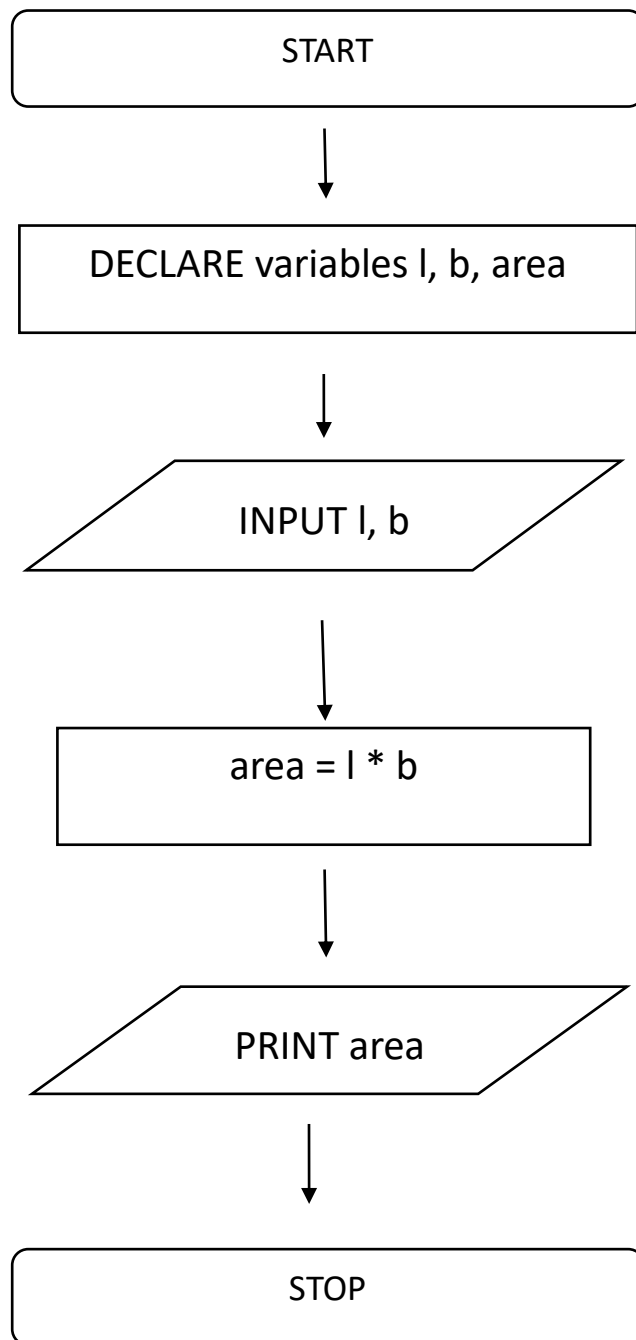
Step 3 : INPUT l, b

Step 4 : $\text{area} = l * b$

Step 5 : PRINT area

Step 6 : STOP

Flowchart :



Code :

```
#include<stdio.h>

#include<conio.h>

void main()
{
    float Length,Breadth;
    clrscr();
    printf("Enter Length Of Rectangle: \n");
    scanf("%f",&Length);
    printf("Enter Breadth Of Rectangle: \n");
    scanf("%f",&Breadth);
    printf("The Area Of Rectangle is: %f",Length*Breadth);
    getch();
}
```

Output :

```
Enter Length Of Rectangle:  
20  
Enter Breadth Of Rectangle:  
17  
The Area Of Rectangle is: 340.000000_
```

(6) Write a program that input 3 values from user and find out its Average.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables x, y, z, avg

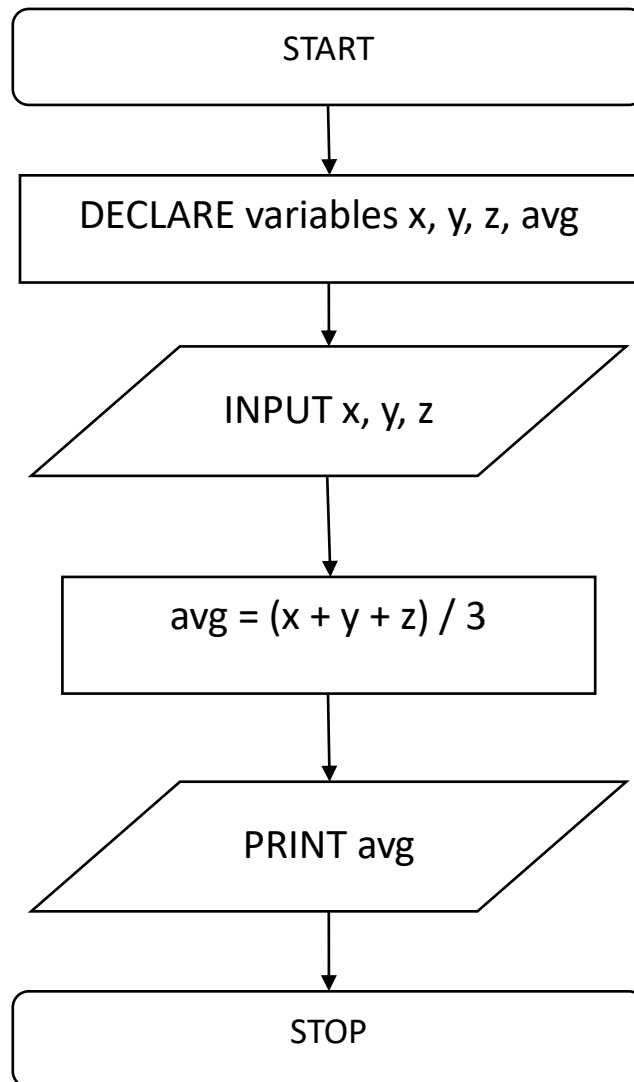
Step 3 : INPUT x, y, z

Step 4 : $\text{avg} = (x + y + z) / 3$

Step 5 : PRINT avg

Step 6 : STOP

Flowchart :



Code :

```
#include<stdio.h>

#include<conio.h>

void main()
{
    int x,y,z;
    clrscr();
    printf("\n Enter Number 1: ");
    scanf("%d",&x);
    printf("\n Enter Number 2: ");
    scanf("%d",&y);
    printf("\n Enter Number 3: ");
    scanf("%d",&z);
    printf("\n Average is: %d ",(x+y+z)/3);
    getch();
}
```

Output :

```
Enter Number 1: 23
```

```
Enter Number 2: 33
```

```
Enter Number 3: 43
```

```
Average is: 33 _
```

(7) Write a program to Calculate the Square and Cube of a number.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables n, sqr, cube

Step 3 : INPUT n

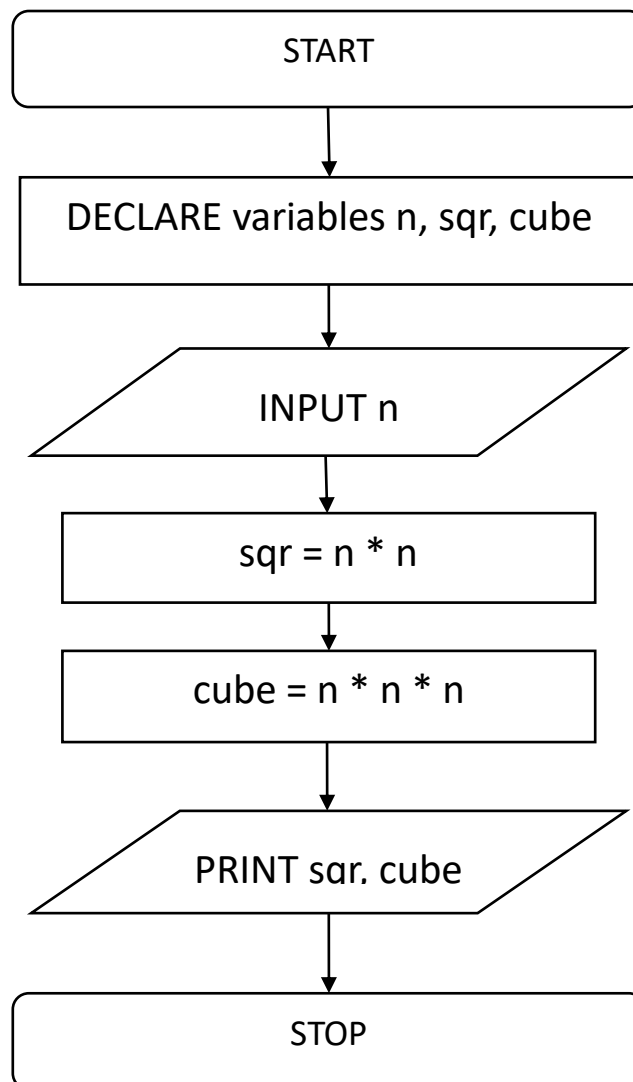
Step 4 : $\text{sqr} = n * n$

Step 5 : $\text{cube} = n * n * n$

Step 6 : OUTPUT sqr, cube

Step 7 : STOP

Flowchart :



Code :

```
#include<stdio.h>

#include<conio.h>

void main()
{
    int n;
    clrscr();
    printf("\n Enter Number: ");
    scanf("%d",&n);
    printf("\n Square is: %d",n*n);
    printf("\n Cube is: %d",n*n*n);
    getch();
}
```

Output :

```
Enter Number: 2
```

```
Square: 4
```

```
Cube is: 8
```

(8) Write a program Interchange the value of two numbers using third variable.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables temp, x, y

Step 3 : INPUT x, y

Step 4 : temp = x

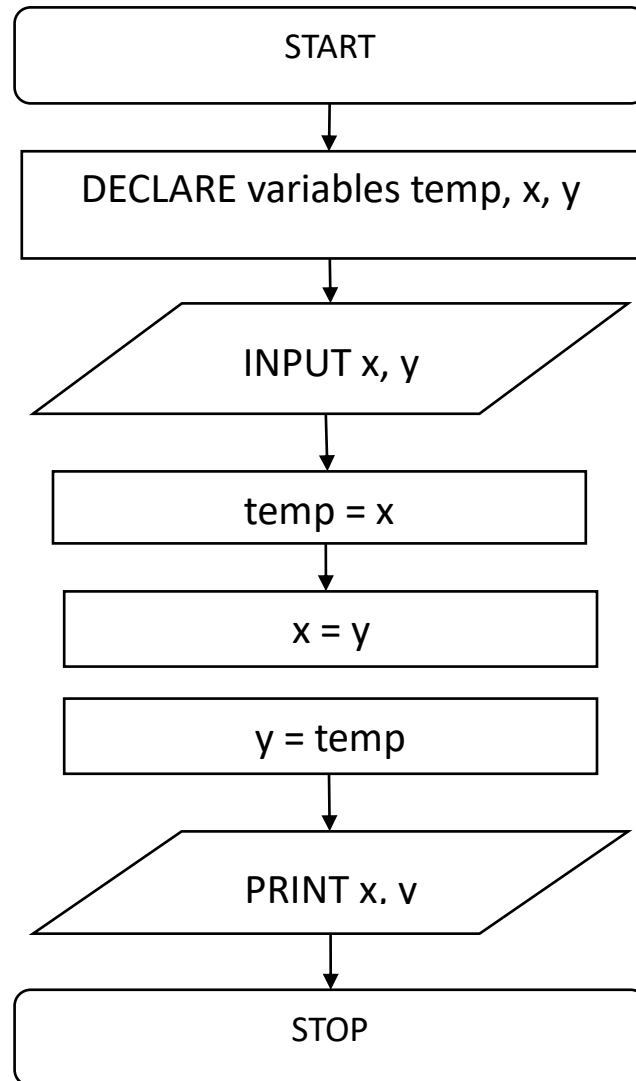
Step 5 : x = y

Step 6 : y = temp

Step 7 : PRINT x, y

Step 8 : STOP

Flowchart :



Code :

```
#include<stdio.h>

#include<conio.h>

void main()
{
    int temp,x,y;
    clrscr();
    printf("\n Enter Number 1: ");
    scanf("%d",&x);
    printf("\n Enter Number 2: ");
    scanf("%d",&y);
    temp=x;
    x=y;
    y=temp;
    printf("\nNew Number 1: %d",x);
    printf("\nNew Number 2: %d",y);
    getch();
}
```

Output :

```
Enter Number 1: 23
Enter Number 2: 33
New Number 1: 33
New Number 2: 23_
```

(9) Write a program Interchange the value of two numbers without using third variable.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables x, y

Step 3 : INPUT x, y

Step 4 : $x = x + y$

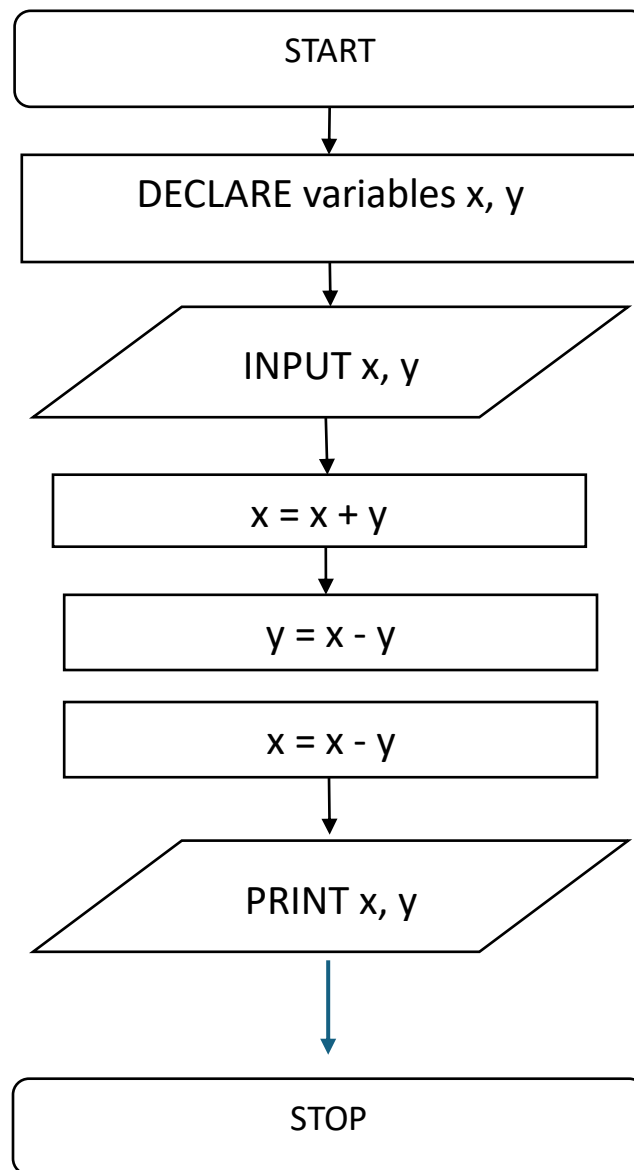
Step 5 : $y = x - y$

Step 6 : $x = x - y$

Step 7 : PRINT x, y

Step 8 : STOP

Flowchart :



Code :

```
#include<stdio.h>

#include<conio.h>

void main()
{
    int x,y;
    clrscr();
    printf("\n Enter Number 1: ");
    scanf("%d",&x);
    printf("\n Enter Number 2: ");
    scanf("%d",&y);
    x=x+y;
    y=x-y;
    x=x-y;
    printf("\nNew Number 1: %d",x);
    printf("\nNew Number 2: %d",y);
    getch();
}
```

Output :

```
Enter Number 1: 23
Enter Number 2: 33
New Number 1: 33
New Number 2: 23_
```

(10) Write a program that input quantity, price, discount and display final amount.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables q, pr, disc, f_amt

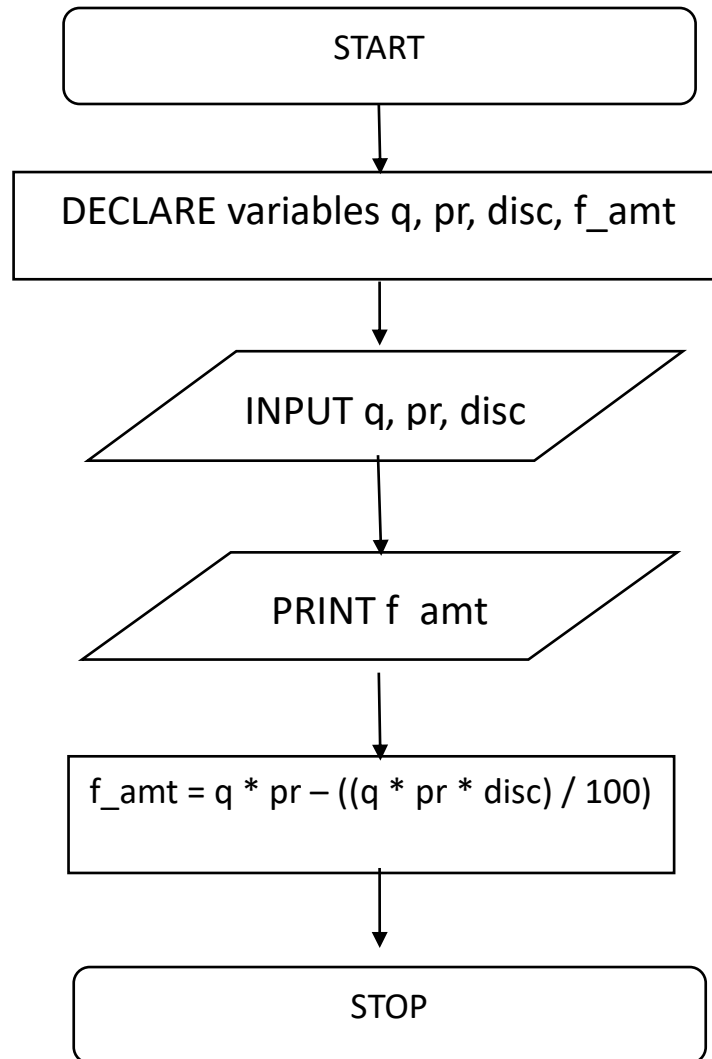
Step 3 : INPUT q, pr, disc

Step 4 : $f_amt = q * pr - ((q * pr * disc) / 100)$

Step 5 : OUTPUT f_amt

Step 6 : STOP

Flowchart :



Code :

```
#include<stdio.h>

#include<conio.h>

void main(){

    float Input_quantity,Price,Discount,Final_Amount;

    clrscr();

    printf("Enter Quantity of Items: \n");

    scanf("%f",&Input_quantity);

    printf("Enter Price Of Item: \n");

    scanf("%f",&Price);

    printf("Enter Discount: \n");

    scanf("%f",&Discount);

    Final_Amount = Input_quantity*Price-
((Input_quantity*Price*Discount)/100);

    printf("The Final Amount Is: %.2f",Final_Amount);

    getch();

}
```

Output :

```
Enter Quantity of Items:  
10  
Enter Price Of Item:  
300  
Enter Discount:  
2  
The Final Amount Is: 2940.00
```